



ASSIGNMENT JUNE SEMESTER 2025

INTRODUCTION TO SUSTAINABLE INFRASTRUCTURE & TECHNOLOGY (UGCS 1001)

SUBMISSION 19th SEPTEMBER 2025

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INSTRUCTION & SUBMISSION

1. **This is a group assignment and should consist of 3 students per group**
2. **Submission Deadline:** On or before **19th September 2025**, not later than **5:00 PM**.
3. **Late Submission Penalty:** A deduction of **2% per day** from the overall marks will be applied for late submissions.
4. **Submission Requirement:** Please attach this **“Cover Page”** to your final submission.

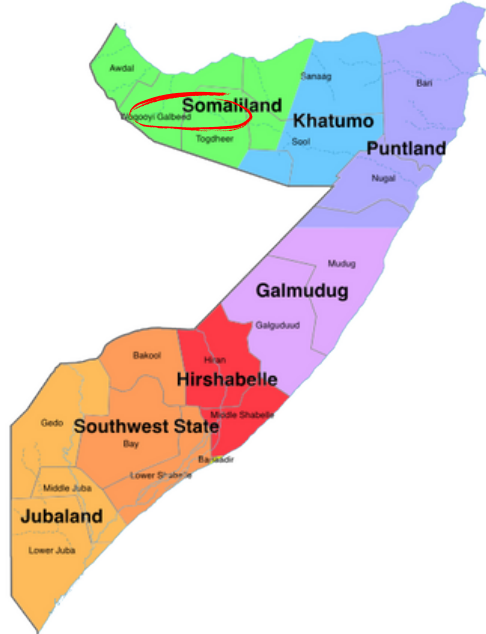


Introduction to Sustainable Infrastructure & Technology (UGCS 1001)

HARGEISA URBAN WATER SUPPLY UPGRADING PROJECT (HUWSUP)

ABSTRACT

The Hargeisa Urban Water Supply Upgrading Project (HUWSUP) is a comprehensive strategy to resolve the critical water crisis in Somaliland's capital. By replacing a **40-year-old** pipeline system that resulted in over **80% water loss**, the project increases water transmission capacity from **9,000 m³/day** to **24,000 m³/day**. The integration of a modernized pumping station, an innovative SCADA control system, and a robust capacity-building program for the Hargeisa Water Agency ensures a secure and sustainable water future, providing over **14,500** previously unserved households with a reliable water source.



INFRASTRUCTURE CLASSIFICATION

The HUWSUP integrates multiple infrastructure types to create a holistic and sustainable system:

- Hard Infrastructure (The Physical Assets):** The tangible components built for the project, including the new **23km, 600mm** ductile iron pipeline, 4 new deep boreholes, and the modernized central pumping station.
- Soft Infrastructure (The Human & System Assets):** The non-physical systems required for the hardware to function effectively, primarily the **"learning-by-doing"** capacity-building program for Hargeisa Water Agency (HWA) staff in modern system operation and maintenance.

INTRODUCTION

Hargeisa's rapid urban growth has completely outpaced its failing infrastructure. The significance of the HUWSUP lies in its direct response to a severe urban crisis that was impacting public health and economic stability. The key challenges were:

- Failing Infrastructure:** A **40-year-old**, undersized pipeline was beyond its service life, causing frequent system failures.
- Catastrophic Water Loss:** Over **80%** of the water supply was lost to leakages along the old pipeline route, wasting a scarce resource in an arid region.
- Inadequate Supply:** The old system's capacity of **9,000 m³/day** was insufficient for a growing capital city.
- Limited Access & Economic Burden:** Over **14,500** households were completely unserved, forcing families to spend up to **20%** of their income on expensive private water vendors.

The project is located in Hargeisa, with a main infrastructure corridor running **23km** from the **Hora Haadley** and **Ged Deeble** wellfields to the main city reservoir. The core construction phase took approximately **5 years**, utilizing a blended workforce of skilled Somaliland technicians and international experts. To ensure a **40+ year lifespan**, **high-grade 600mm** ductile iron pipes were procured from a leading international manufacturer, and a strict Health, Safety, and Environment (HSE) plan was enforced.



SUSTAINABILITY & INNOVATION FEATURES

Sustainability & Climate Resilience:

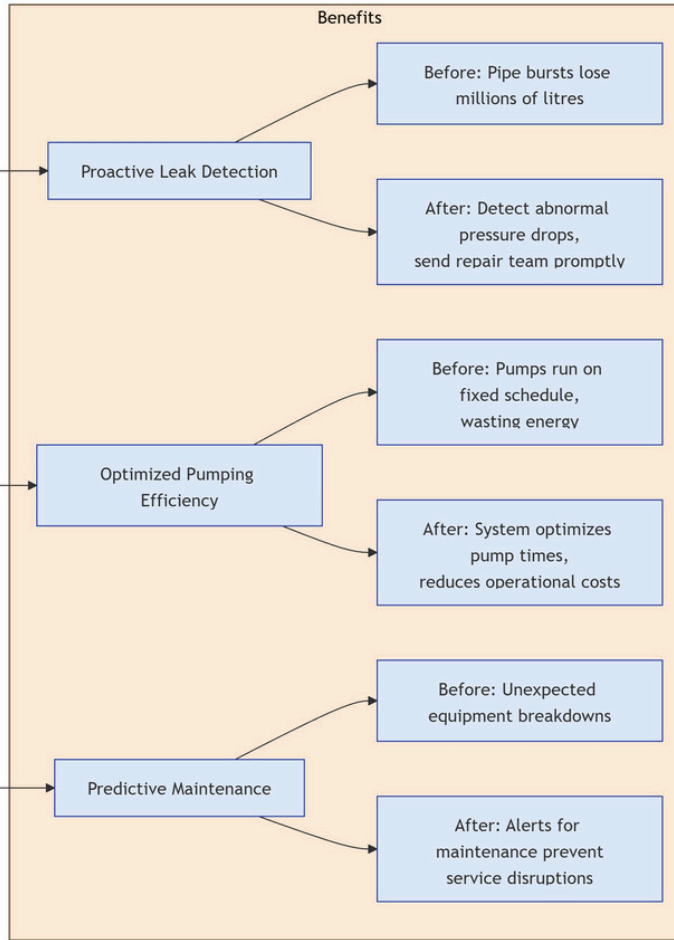
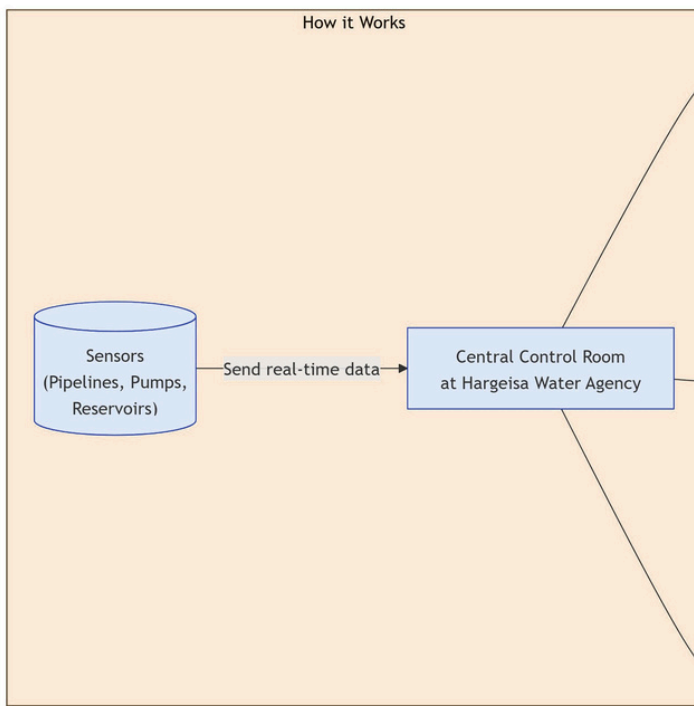
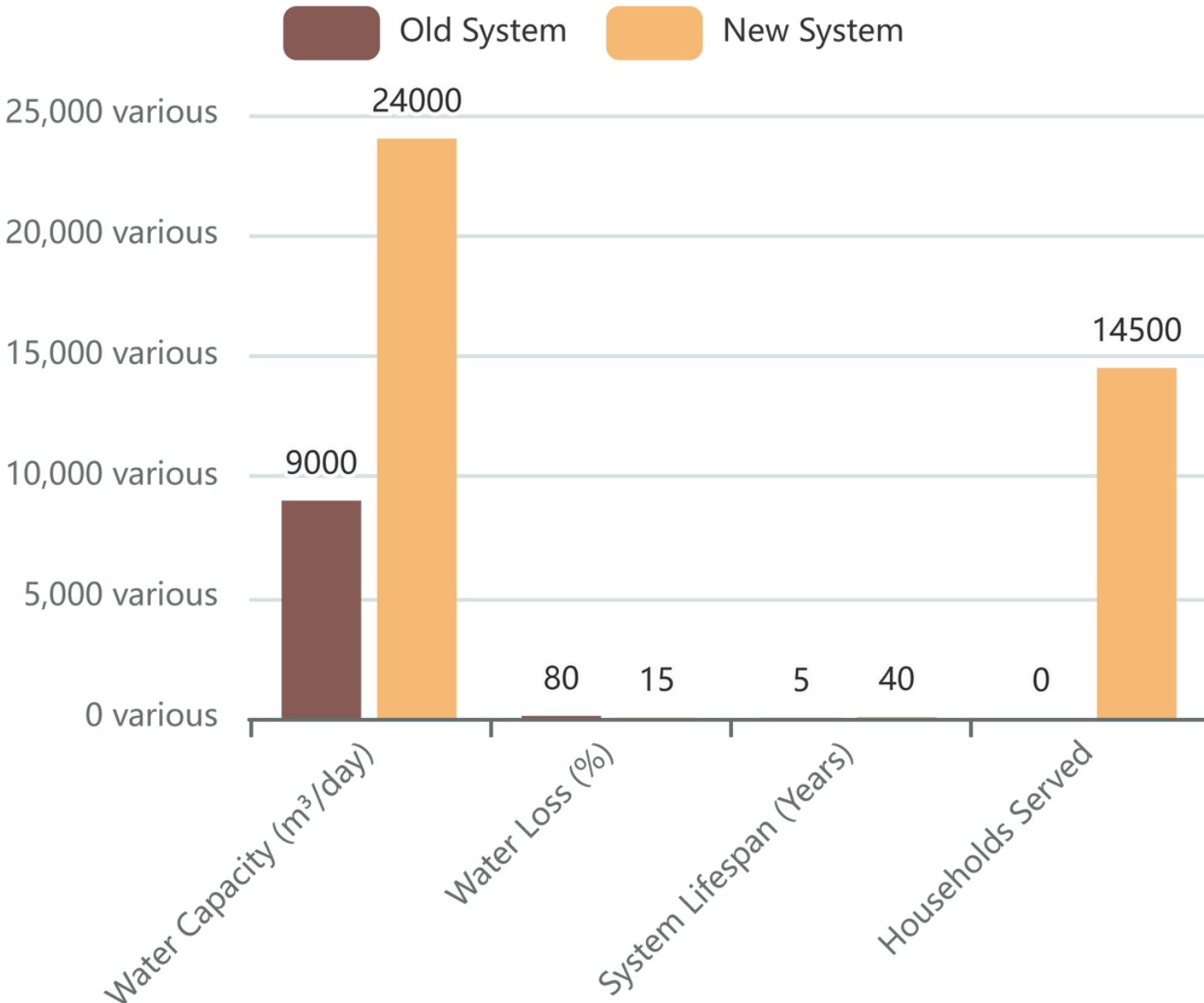
- Durability:** The use of ductile iron pipes and modern construction increases the system's lifespan from **5 years** to over **40 years**.
- Drought Resilience:** Developing deeper boreholes creates a drought-resilient water source, a critical climate adaptation measure.

Technology Adoption (Innovation):

- The project's key innovation is the Supervisory Control and Data Acquisition (**SCADA**) system. This IoT-based system uses sensors to send real-time data on water flow and pressure to a central control room, enabling proactive leak detection, optimized pumping efficiency, and predictive maintenance.

IMPACT ASSESSMENT

Social Impact Assessment (SIA): The project yields profound social benefits by providing more than **14,500** previously unserved households with access to a safe and reliable water source. This directly reduces waterborne diseases, improves sanitation, and significantly reduces the economic and time burden on families, especially women and girls, freeing up resources for education and health.



OBSERVATION & FURTHER IMPROVEMENT

Observation (Strengths & Limitations):

The HUWSUP is an exemplary case of replacing critical infrastructure. Its key strength is the integrated approach, combining durable **Hard Infrastructure** with essential capacity-building in **Soft Infrastructure**. A limitation was the necessary, but temporary, reliance on diesel generators before the full renewable energy transition.

Further Improvement (Recommendations):

- Implement City-Wide Smart Metering:** Install smart meters at the household level to manage demand and reduce waste within the city network.
- Accelerate Full Solar Transition:** Secure funding to build a full **1.5 MW solar farm**, eliminating reliance on diesel and making the system **100%** energy-sustainable.
- Launch a Water Conservation Program:** Develop a public awareness campaign with HWA to promote responsible water use.

